THE CATEGORY OF ABELIAN SHEAVES ON A SITE IS ABELIAN, WITH KERNEL AND COKERNEL DESCRIBED BY PRESHEAF OPERATIONS AND SHEAFIFICATION

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Allowed standard prerequisites (editorial).

- Adjoint functors yielding an abelian structure on the target category. Stacks Project, Homology, Lemma homology-lemma-adjoint-get-abelian (tag 010B). The general criterion applied to the inclusion and sheafification pair, from which abelianness and the kernel/cokernel descriptions follow.
- Sheafification of abelian presheaves: adjointness and left exactness. Stacks Project, Sites, Section sites-section-sheaves-algebraic-structures (tag 00YR). Supplies the properties of sheafification that the criterion's hypotheses require.

Source reference. Original reference 010B: sources/stacks-chow/tags.txt, source lines 1321–1321. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00V6: sources/stacks-chow/tags.txt, source lines 1136–1136. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00WM: sources/stacks-chow/tags.txt, source lines 1188–1188. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03A3: sources/stacks-chow/tags.txt, source lines 4294–4294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original section 03A6: sources/stacks-chow/tags.txt, source lines 4297–4297. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of abelian sheaves on $\mathcal{C}$.*

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(1) *The category $Ab(\mathcal{C})$ is an abelian category.*
(2) *The kernel* $\mathrm{Ker}(\varphi)$ *of φ is the same as the kernel of φ as a morphism of presheaves.*
(3) *The morphism φ is injective (Homology, Definition 010B) if and only if φ is injective as a map of presheaves (Sites, Definition 00V6), if and only if φ is injective as a map of sheaves (Sites, Definition 00WM).*
(4) *The cokernel* $\mathrm{Coker}(\varphi)$ *of φ is the sheafification of the cokernel of φ as a morphism of presheaves.*
(5) *The morphism φ is surjective (Homology, Definition 010B) if and only if φ is surjective as a map of sheaves (Sites, Definition 00WM).*
(6) *A complex of abelian sheaves*
$$\mathcal{F} \to \mathcal{G} \to \mathcal{H}$$
is exact at $\mathcal{G}$ if and only if for all $U \in \mathrm{Ob}(\mathcal{C})$ and all $s \in \mathcal{G}(U)$ mapping to zero in $\mathcal{H}(U)$ there exists a covering $\{U_i \to U\}_{i \in I}$ in $\mathcal{C}$ such that each $s|_{U_i}$ is in the image of $\mathcal{F}(U_i) \to \mathcal{G}(U_i)$.

Proof. We claim that Homology, Lemma 03A3 applies to the categories $\mathcal{A} = Ab(\mathcal{C})$ and $\mathcal{B} = PAb(\mathcal{C})$, and the functors $a : \mathcal{A} \to \mathcal{B}$ (inclusion), and $b : \mathcal{B} \to \mathcal{A}$ (sheafification). Let us check the assumptions of Homology, Lemma 03A3. Assumption (1) is that $\mathcal{A}$, $\mathcal{B}$ are additive categories, a, b are additive functors, and a is right adjoint to b. The first two statements are clear and adjointness is Sites, Section 00YR (ϵ). Assumption (2) says that $PAb(\mathcal{C})$ is abelian which we saw in Section 03A6 and that sheafification is left exact, which is Sites, Section 00YR (ζ). The final assumption is that $ba \cong \mathrm{id}_{\mathcal{A}}$ which is Sites, Section 00YR (δ). Hence Homology, Lemma 03A3 applies and we conclude that $Ab(\mathcal{C})$ is abelian.

In the proof of Homology, Lemma 03A3 it is shown that $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ are equal to the sheafification of the kernel and cokernel of φ as a morphism of abelian presheaves. This proves (4). Since the kernel is a equalizer (i.e., a limit) and since sheafification commutes with finite limits, we conclude that (2) holds.

Statement (2) implies (3). Statement (4) implies (5) by our description of sheafification. The characterization of exactness in (6) follows from (2) and (5), and the fact that the sequence is exact if and only if $\mathrm{Im}(\mathcal{F} \to \mathcal{G}) = \mathrm{Ker}(\mathcal{G} \to \mathcal{H})$. □